WELLSENSE · DESIGN DECISION RECORD

WellSense concept selection

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DOCUMENT CONTROL

Document ID	WLS-DDR-005	Version	0.1
Status	Draft	Date	2026-10-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsense		
Source	docs/decisions/0005-concept-selection.md @ 9c67a3d		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Concept selection between options A, B and C of WLS-OPT-001 argued; A and C within 10 %, both put to Amish without a pick; proposed, awaiting Amish

0005: Concept selection

- **Date:** 2026-10-25
- **Status:** proposed, awaiting Amish. Nothing here is decided; the repo stays on option A, the current design, until Amish decides (STANDARDS section 21, existing designs).

Context

Kit 1.8 asks every repo for at least two concept options with different working principles, scored against the same user requirements (STANDARDS section 21), and kit 1.9 asks for every option to be modelled to the repo's detail level (D3 at P3; Amish, 2026-10-24: "Detail every option"). WellSense was designed before kit 1.8, so its current design is option A. Two other ways of turning a water level into a number fit the same wellhead, tube, post, junction box and FieldNode core: a bubbler (B) and an acoustic echo sounder (C). All three are in `cad/src/model.py` (`OPTIONS`, `components(option)`) and pass the D3 fidelity check (`cad/fidelity.json`); their STEM rows are in WLS-STM-001 and their scoring in WLS-OPT-001.

Options

Table 1. The options and their weighted scores (WLS-OPT-001, Table 2; of 215).

Option	Working principle	Score	Cost at 30 m (USD)	Sensor energy	Main open point
A, vented transducer in an access tube (current design)	Hydrostatic head on a vented 4 to 20 mA probe hung in the tube	173	264.40 (priced)	38.2 mWh/ day	Calibrated accuracy 37.1 mm worst case (STM S7); drift (S10); certificates for the probe and cable (PR-009)
B, bubbler with a surface sensor	Back-pressure of air purged down a PE line, read at the surface	147	about 233 (estimate)	263 mWh/ day	Energy 11 % of FieldNode's allowance against 5 % (PR-008); pump and valve wear; pump pressure (B4)
C, acoustic echo sounder	Time of flight of a chirp down the tube's air column to the water surface	164	about 205 (estimate)	10.6 mWh/ day	Speed-of-sound calibration from joint echoes unproven in PVC (C2); signal at depth (C3); condensation on the head (C5)

Recommendation

Options A and C score within 10 % of each other (164 is 5.2 % below 173), so both go to Amish without a pick, as STANDARDS section 21 requires. For his choice:

- **Choose A** to keep the proven, fully drawn and priced design; its open points are evidence (a side-by-side tape record and supplier certificates), not principle.
- **Choose C** to pursue a cheaper logger with nothing new in the water and a quarter of A's energy, accepting that its accuracy depends on an echo calibration that needs a TRL 4 echo test in a real PVC tube before any redesign of the drawing, BOM or build plan.

Option B is not put forward: its pump energy misses PR-008 and it adds wear parts, for an accuracy no better than A's.

Graft for either choice: record the depth of every tube joint on the build sheet (C's idea), so a later acoustic check head or a tape can confirm the probe depth without pulling it.

Consequences

- If Amish chooses A: `project.yaml` `concept_chosen: A`; no design change.
- If Amish chooses C: `CHOSEN` in `cad/src/model.py` stays A until the TRL 4 echo test is planned and approved; the C head, ranging board and firmware get their own STEM closure, BOM lines, drawing and build plan in a later session.
- Either way the decision is recorded in WLS-DEC-001 and WLS-PRC-001 is updated to describe the chosen option.